

The Face and Future of Cyberthreats: Emerging Challenges and Trends

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Abstract: The cybersecurity landscape is rapidly evolving, driven by emerging technologies and increasingly sophisticated threats. This systematic literature review synthesizes challenges, defensive strategies, and sector-specific vulnerabilities identified across 52 scholarly sources published between 2015 and 2025. Artificial Intelligence (AI) and machine learning are highlighted as both critical threat vectors and promising defensive mechanisms, featuring in nearly half of the studies. Human factors, including skills gaps, human error, and training deficiencies, remain pervasive vulnerabilities, cited in over 35% of the literature. The review identifies finance, government, and healthcare as the most frequently impacted sectors, while emerging threats such as ransomware, supply chain attacks, and phishing continue to escalate. Geographically, research is concentrated in India and the USA, indicating potential limitations in the global generalizability of findings. Foundational strategies, including security frameworks, standards, and awareness programs, remain prominent in proposed mitigations. This review underscores the need for balanced attention to underrepresented threats and enhanced methodological transparency, providing an evidence-based foundation to guide researchers, industry stakeholders, and policymakers in developing proactive and resilient cybersecurity strategies. The cybersecurity landscape is rapidly evolving as emerging technologies expand attack surfaces and adversaries escalate in sophistication. To synthesize emerging challenges, sectoral impacts, and defensive trends in cybersecurity and identify evidence gaps. Google Scholar, Scopus, and Web of Science; publications 2015–2025. Study eligibility criteria: Peer-reviewed, English-language studies addressing emerging cyberthreats or defensive paradigms, with an explicit research framework/methodology. Appraisal and synthesis methods: Independent dual/triple screening; data extraction by three reviewers with consensus; quality appraisal using JBI and Newcastle–Ottawa Scale; narrative/thematic synthesis. 52 studies met criteria (journals 57.69%, conferences 25.00%). Research is geographically concentrated (India 23.08%, USA 21.15%). AI/ML appears as both threat and defence in ≈ 27 –31% of studies; human factors are cited in $\geq 35\%$. Most-studied sectors: finance and government (each 13.46%); energy/health/transport (5.77% each). Common threats: AI/ML-driven attacks, industrial/critical-infrastructure risks, cloud/supply-chain compromise. Dominant defences: AI-driven detection/response, security hygiene, and training/awareness. Conceptual bias (limited empirical validation), sector/region skew, and potential reporting/selection bias. Evidence supports integrated, proactive cyber defence combining AI-enabled controls with governance and human-centred measures. Gaps include post-quantum preparedness, supply-chain resilience, and coverage of under-studied regions/sectors.

Keywords: cybersecurity; emerging threats; AI/ML; critical infrastructure; systematic review; defence strategies.

1. Introduction

Cybersecurity has become one of the most pressing challenges of the digital age [1]. Rapid adoption of AI, IoT, and cloud computing has expanded the attack surface, creating new opportunities for cybercriminals [2]. At the same time, AI features prominently on both sides of the equation—powering defences and enabling novel attacks [3]. These dynamics elevate risks to critical infrastructure, economies, and societies [4]. Emerging work argues for a shift from reactive to predictive security using trend analysis and AI/ML [5], while human factors—like developer awareness and secure practices—remain decisive [6]. Domain-specific surveys (e.g., blockchain) catalogue concrete challenge taxonomies that highlight unresolved gaps [7]. Forward-looking syntheses integrate trends such as AI, Zero Trust, and quantum-era considerations [8]. Concurrently, broad reviews continue to chart the growth and pervasiveness of cybercrime at a high level [9]. Against this backdrop, this review addresses the

lack of a holistic, cross-domain synthesis connecting emerging threats with defensive trends. Using a systematic literature review of studies from 2015 to 2025, it provides a unified perspective on the evolving landscape. It aggregates scattered findings to answer key questions about emerging threats and the effectiveness of contemporary countermeasures—including AI-driven solutions and proactive frameworks like Zero Trust [8]—and identifies gaps that demand intelligent, proactive, and collaborative defence mechanisms capable of keeping pace with sophisticated adversaries [8]. Table 1 presents a comparative analysis of key review studies on emerging challenges and trends in cybersecurity [1–10], along with the proposed systematic review.

Table 1. Comparative Analysis of the Existing Review Works and Proposed Systematic Review on the Emerging Challenges and Trends in Cybersecurity.

Ref.	Contribution	Pros	Cons
[1]	Provides a broad conceptual framework linking cyber-threats to national security and economic stability.	Establishes a strong foundational understanding of the strategic impact of cyber-incidents.	Lacks empirical data and detailed technical analysis of specific attack vectors.
[2]	Emphasizes the societal perception and human factors in cybersecurity, moving beyond a purely technical focus.	Clearly links the evolution of threats to the increasing connectivity of IoT and critical systems.	The findings are based on "perception," which may introduce subjectivity rather than empirical data.
[3]	Provides a focused examination of AI's dual role as a transformative defence tool and a source of new challenges (e.g., ethical concerns, adversarial attacks).	Offers a clear, structured overview of the cybersecurity landscape and the historical evolution of threats and AI's role.	The analysis remains at a high level, lacking in-depth technical specifics on how AI models are implemented or tested.
[4]	Proposes a holistic mitigation framework that combines technology, employee/consumer training, and policy/regulation.	Focuses on a highly relevant and targeted domain (e-commerce), making the analysis practical for businesses in that sector.	The scope is limited to four threat types (social engineering, DoS, malware, data attacks), omitting others like supply-chain or API vulnerabilities.
[5]	Advocates for a strategic shift from reactive to predictive cybersecurity using trend analysis and AI/ML.	Correctly identifies critical challenges for predictive security, such as ethical concerns like data bias and the complexity of the threat landscape.	The analysis remains theoretical and conceptual, without presenting new data, models, or empirical validation of the predictive approach.
[6]	Addresses the critical human factor in cybersecurity by focusing on developer awareness and training, moving beyond purely technical solutions.	Presents an empirical study with real-world evaluation through four industry events and three surveys, providing tangible evidence of effectiveness.	The platform and study are highly specific to the C/C++ programming language, limiting immediate applicability to other widely used languages.
[7]	Systematically categorizes 30 distinct cybersecurity challenges in Blockchain Technology (BCT) into six standardized classes,	Highlights that most literature discusses challenges in general terms, clearly defining the scope of existing knowledge and its limitations.	Identifies a significant gap (lack of sector-specific strategies) but does not attempt to fill it, leaving the practical application unresolved.

	providing a clear taxonomy.		
[8]	Synthesizes key future trends (AI, Zero Trust, quantum computing) into a single forward-looking framework.	Covers a wide range of modern concepts, from ransomware evolution to regulatory governance.	The analysis is entirely conceptual, offering no new data, case studies, or empirical validation.
[9]	Identifies the dual focus on new technologies as both a source of threats and a means for security solutions.	Clearly states the pervasive and growing nature of cybercrimes.	The analysis is very high-level and lacks depth, specific examples, or detailed exploration of the trends mentioned.
	It provides guidance for future studies and practice.	High-quality studies were systematically included.	Most studies focus on a few countries and sectors. This limits the global generalizability of findings.

As synthesized in Table 1 [1–9], prior reviews collectively illuminate but do not integrate the cybersecurity landscape: many are domain-bound (national strategy, sectoral frameworks, or blockchain taxonomies) and thus fragment technical, human, and policy lenses; a substantial share advance conceptual or forward-looking models without empirical validation; sector-specific emphases constrain cross-industry and cross-national generalizability; human and organizational factors, while acknowledged, remain insufficiently embedded alongside technical controls; predictive, AI-driven, and proactively preventive approaches are underrepresented relative to reactive detection/response; and even broad syntheses lack actionable implementation guidance and metrics. These gaps motivate a unified, empirically grounded, and globally inclusive synthesis that links emerging technologies, human/organizational practices, and proactive paradigms to operationalize resilient cyber defence.

1.1. Research questions

This systematic review set out to understand the rapidly evolving field of cybersecurity by examining where current research is concentrated, what key challenges have been highlighted, and which solutions are being proposed. The goal is to identify patterns in existing studies, expose gaps in knowledge, and provide direction for future work that can improve defences across different sectors and regions. To achieve this, the review was guided by the following research questions:

- How has cybersecurity research grown over the past decade, and what does this reveal about the level of attention given to emerging threats?
- Which countries and regions are leading in cybersecurity research, and what gaps exist in geographic coverage?
- What are the most common cybersecurity challenges and threat categories identified in the literature, and how are they changing over time?
- Which defensive strategies, technological, human, and governance-related, are most emphasized, and how effective are they considered to be?
- How are different sectors being addressed in cybersecurity research, and which industries remain underexplored?
- What future trends and directions do researchers anticipate in cybersecurity, and what is the level of certainty around these predictions?

1.2. Rationale

The rationale for this systematic review is to address the fragmented and uneven nature of existing cybersecurity research. Although numerous studies exist, most focus narrowly on specific technologies; particularly AI and machine learning or originate from a small number of countries such as India and the USA. This leaves other regions, industries, and defensive strategies underrepresented. With the rise of new risks such as supply chain compromises, IoT vulnerabilities, and sector-specific threats in areas like healthcare and energy, there is a clear need for a comprehensive synthesis. This review therefore consolidates scattered findings, highlights gaps, and provides insights into the emerging challenges and defensive trends shaping the future of cybersecurity.

1.3. Objectives

This systematic review aims to provide a clear and comprehensive understanding of the evolving cybersecurity landscape, focusing on both emerging threats and the strategies developed to mitigate them. The review seeks to synthesize existing research to uncover patterns, identify gaps, and support evidence-based decision-making for academia, industry, and policymakers. Specifically, the review has the following objectives:

- To identify and analyse the most significant emerging cyberthreats driven by technological change, including AI, IoT, and quantum-related risks.
- To examine the defensive strategies and solutions being developed across sectors to address these threats.
- To synthesize insights from existing research to guide future academic studies, industrial practices, and policymaking in cybersecurity

1.4. Research contributions

This systematic review makes several key contributions to the field of cybersecurity:

- Synthesizes key threats, including AI/ML attacks, IoT vulnerabilities, and critical infrastructure risks.
- Highlights AI-driven defences, security hygiene, human training, and governance frameworks.
- Shows research focus in finance and government sectors, and regional disparities in cybersecurity studies.
- Reveals underexplored areas such as quantum threats, agriculture, smart grids, and less-studied regions.
- Offers actionable insights for academia, industry, and policymakers to strengthen cyber resilience.

1.5. Research novelty

This study is among the first to provide a comprehensive, cross-cutting review of emerging cyberthreats and defensive trends across multiple technological domains. Unlike previous reviews that focus on specific threats, technologies, or industries, this systematic review synthesizes the latest research from 2015 to 2025 to offer a holistic view of the evolving cybersecurity landscape, identifying underexplored threats, gaps in current knowledge, and actionable insights for academia, industry, and policymakers.

2. Materials and Methods

In this subsection, the study outlines the methodology employed to conduct a systematic review focusing on the emerging cyberthreats and defensive trends in cybersecurity. The review covers literature published over the last decade, from 2015 to 2025. To the best knowledge of the authors, no similar comprehensive review exists within this specific timeframe, making this study a novel contribution to the field. The research methodology includes

the careful selection of applicable peer-reviewed articles from key online databases, namely Scopus, Google Scholar, and Web of Science, ensuring a thorough examination of the subject matter [10 -20].

2.1. Eligibility criteria

A systematic study of all peer-reviewed and published research works relevant to the study the merging challenges and trends in cybersecurity was conducted for examination. Only research works published in English between 2015 and 2025 were included in the analysis. A proper criterion for inclusion was adapted to ensure that only studies directly addressing the research topic were considered, while irrelevant works were. Consequently, only peer-reviewed research works that fundamentally focus on the emerging challenges and trends in cybersecurity and that include a research framework or methodology specific to these aspects, were considered. The inclusion and exclusion criteria for this study are tabulated as in Table 2 [21 -25].

Table 2. Proposed Inclusion and Exclusion Criteria.

Criteria	Inclusion	Exclusion
Topic	Article papers focusing on emerging challenges and trends in cybersecurity	Article papers not focusing on emerging challenges and trends in cybersecurity
Research Framework	Articles must include a clear research framework	Articles must that lack a framework or methodology
Language	Must be written in English	Articles published in languages other than English
Period	Articles between 2015 to 2025	Articles outside 2015 and 2025

2.2. Information sources

A systematic search of online databases was conducted to identify relevant studies for this review. The databases Google Scholar, Scopus, and Web of Science were used due to their comprehensive coverage of peer-reviewed literature in the field of cybersecurity and emerging cyberthreats. Each database was thoroughly searched using a combination of keywords related to the study topic, ensuring that the most pertinent research articles were captured. Scopus gave access to a broad range of scientific journals and conference papers, while Google Scholar enabled the inclusion of grey literature and dissertations that might not be indexed elsewhere. Web of Science was used to cross-search and ensure the robustness of the selected studies by providing citation data and impact factors of the journals. The search results from these databases provided the foundation for the literature review, such that it became a balanced and comprehensive set of research items [26 -30].

2.3. Search strategy

The literature for this research was collected from reputable online research databases, focusing on keywords that address both the technological and contextual aspects of emerging cyberthreats and cybersecurity trends. The inclusion of terms such as “emerging trends,” “future challenges,” “risk,” and “threat landscape” ensured the retrieval of studies addressing diverse dimensions of cybersecurity and cyberthreats. A thorough search was carried out in three main repositories: Google Scholar, Scopus, and Web of Science. To find the most relevant studies, a specific set of keywords was utilized. These keywords were: ("cybersecurity" OR "cyber threats" OR "cybercrime" OR "information security") AND ("emerging trends" OR "future challenges" OR "risk" OR "threat

landscape") AND ("systematic review" OR "literature review"). This combination of terms was chosen to guarantee that the search captured studies directly related to the research topic. The search focused on papers published between 2015 and 2025. This time frame was selected to provide the latest and relevant overview of the topic. The search results included 21 200 papers from Google Scholar, 2437 papers from Scopus, and 169 papers from Web of Science. After collecting these papers, they were carefully reviewed and filtered to select only those that were most relevant to the research questions. This process helped to narrow down the literature to the most useful and high-quality sources for this study. Table 3 shows the list of online repositories that were used as well as the total number of results achieved before the initial screening. The Bibliometric Analysis of Study Search Keywords is illustrated in Figure 1 [31-40].

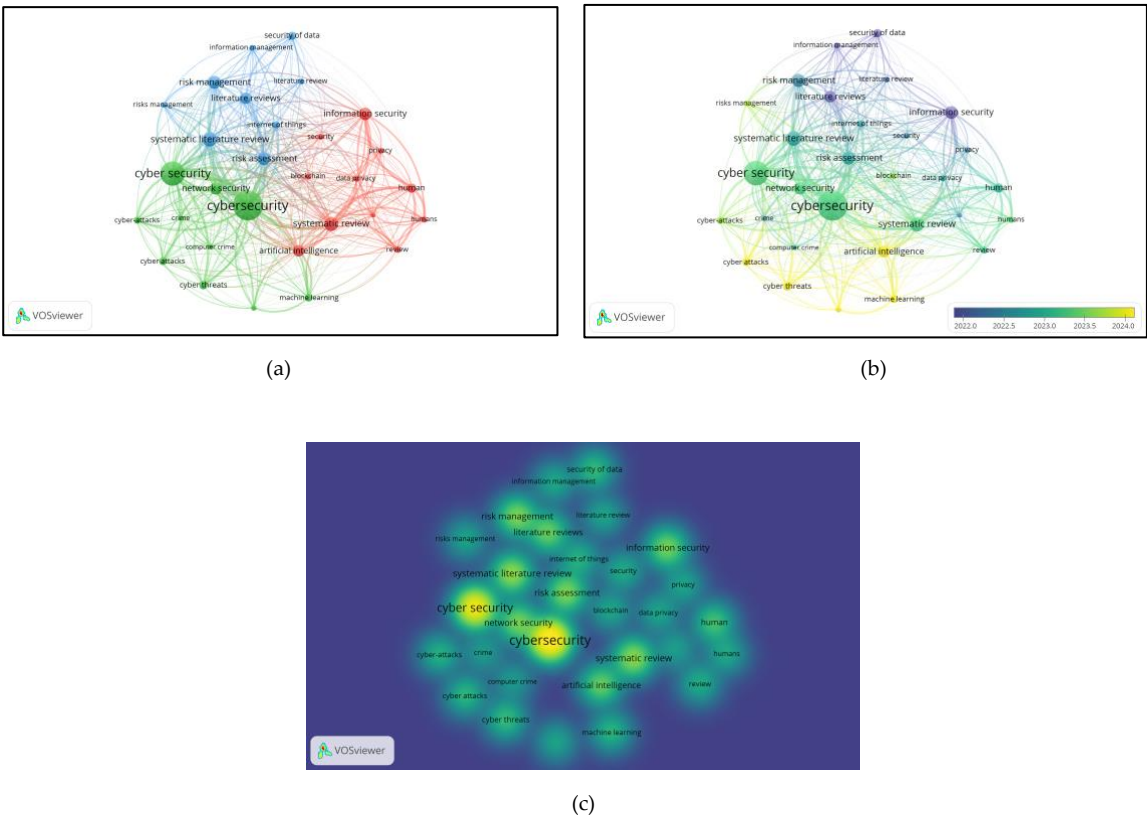


Figure 1. Bibliometric Analysis of Study Search Keywords: (a) Overlay Visualization. (b) Network Visualization. (c) Density Visualization.

Table 3. Results Achieved from Literature Search.

No.	Online Repository	Number of results
1	Google Scholar	21 200
2	Web of Science	169
3	Scopus	2437
Total		23 806

2.4. Selection process

To maximize the relevance of the searched articles, three researchers independently searched Google Scholar, Scopus, and Web of Science. The titles and abstracts of the results were carefully reviewed, and, if necessary, introductions were searched for additional context. A set of keywords, along with Boolean operators (AND, OR), were applied in all three databases to enhance the search process. Moreover, the exclusion and inclusion criteria were employed in a methodical manner to eliminate the irrelevant studies. The key information from the selected studies were summarized and organized in an Excel table [41-43].

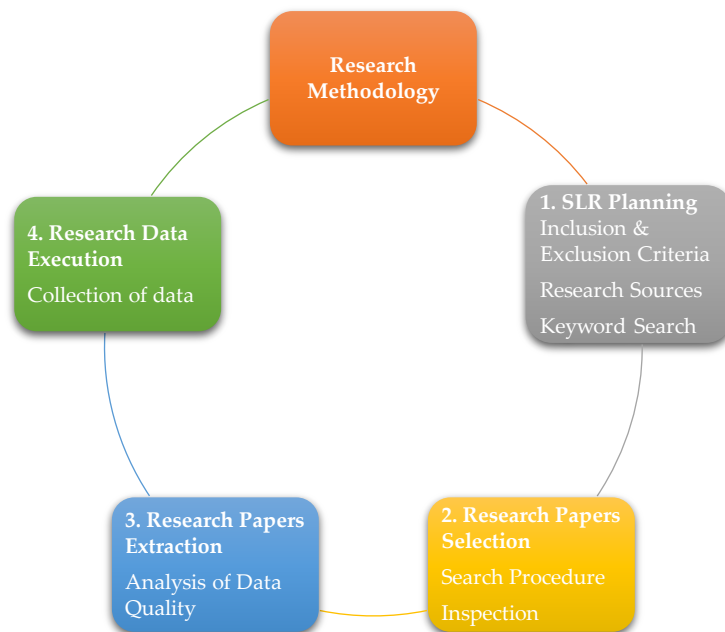


Figure 2. Procedures and Stages of the Review.

2.5. Data collection process

To have a guarantee on the reliability of the data collected, we employed a systematic method that would assist in minimizing errors and bias. Three reviewers independently extracted data from each study under the supervision of a fourth reviewer. Differences in data extracted were resolved in discussion until a consensus was reached. The three reviewers agreed on the final data that will be used for the systematic review. No automation tools were used during this process; instead, all information was entered and double-checked manually to prevent errors. The extracted data was summarized on to Excel, and the fourth reviewer was consulted to confirm the reliability of the data interpretation. For studies with multiple reports, clear selection criteria were applied, with a preference for the most recent and comprehensive publications between 2015 and 2025. Where differences existed among reports, methods and results were critically compared to resolve differences. To maintain uniformity and avoid misinterpretations due to language barriers, research articles published in English were included, while those published in other languages were not. The whole process is illustrated in Figure 3.

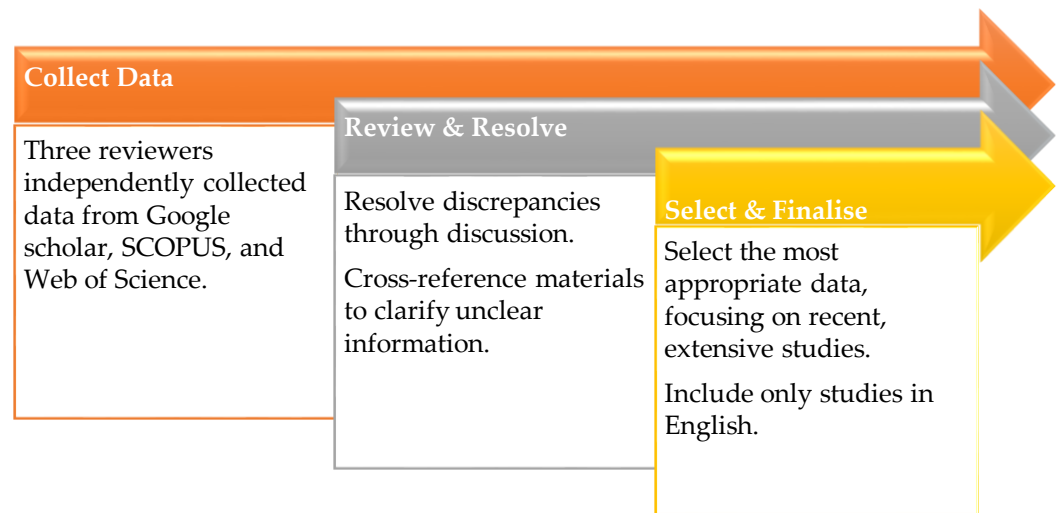


Figure 3. Flow of Data Selection and Extraction.

2.6. Data items

This section provides a comprehensive overview of the data items sought in this systematic review, focusing on both primary outcomes and additional variables relevant to the evolving landscape of cyberthreats and emerging cybersecurity challenges. The primary outcomes encompass various dimensions such as threat frequency, attack sophistication, vulnerability types, defence effectiveness, and organizational impact. In addition to these outcomes, the review also considers study and organizational characteristics, threat actor profiles, technological factors, economic impacts, and external influences, ensuring a thorough contextual understanding of the application and effects of cybersecurity measures across different environments. This approach allows for a nuanced analysis of how emerging cyberthreats contribute to the evolving cybersecurity landscape across diverse settings and conditions.

2.6.1 Data Collection Method

This study primarily measures the impact of cybersecurity, particularly the emerging challenges and threats in cybersecurity. The study aims at capturing the effects of both emerging and currently existing challenges on various domains such as AI & Machine Learning, Governance, Quantum Security, Threat Intelligence & Detection, etc. The inclusion criteria for sorting data were based on the relevance to cybersecurity as outlined in the search strategy using Boolean and proximity operators. Once relevant data was selected, it was sorted for extraction using specific headings to organize the information from various sources. The table below outlines the data collected from the literature reviews.

Table 4. Data Items Collected from Literature Reviews.

Data Items	Description
Title	The title of research related to challenges and trends in cybersecurity
Year	The year the literature was published
Geographic Location	The geographical area where the study was conducted or data was collected
Research Type	The type of research publication focused on emerging challenges and trends in cybersecurity
Online database	The repositories used for collecting literature (Google Scholar, Scopus, Web of Science).
Threat Landscape	The type of cybersecurity threats being analysed in the study (e.g., AI-driven, IoT-based, state-sponsored).
Notable Trends	Key recurring patterns or emerging attack strategies identified in the study.
Sector Impacted	The industry or sector (e.g., healthcare, finance, government, critical infrastructure) is affected by cybersecurity threats.
Emerging Defensive Paradigms	Innovative or next generation defence strategies discussed in the research.
Challenges	Specific challenges in cybersecurity, such as identifying attackers (attribution) or coping with fast-moving threats.
Forecasted Evolutions	The predicted future developments in cyber threats and defence mechanisms.
Future Research Directions	Suggested areas where further academic or practical research is needed to address cybersecurity challenges.

Efforts were made to ensure a comprehensive understanding of the impact of emerging cyberthreats and cybersecurity challenges on organizations, and we thoroughly identified and defined relevant outcomes that capture the technical, operational, and strategic dimensions influenced by these evolving threats. Our approach was designed to synthesize robust evidence that reflects the transformative effects of cyberthreats in the modern digital landscape.

2.6.2 Definition of Collected Data Variables

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approach was designed to synthesize robust evidence that reflects the transformative effects of cyberthreats in the modern digital landscape.

The primary outcomes of this systematic review centred on several key domains that directly relate to emerging cyberthreats and cybersecurity challenges. Threat Evolution was a major outcome, defined by measuring changes in attack vectors, sophistication levels, and frequency patterns. We sought all results that could reflect how cyberthreats have evolved, including the emergence of AI-driven attacks, quantum computing threats, and IoT-specific vulnerabilities. These metrics provided clear insights into the dynamic nature of the threat landscape and the pace of cyberthreat evolution.

Vulnerability Assessment was another critical outcome, assessed by tracking the identification and classification of new vulnerabilities across different systems and sectors. By quantifying vulnerability types, severity levels, and exploitation rates, this outcome provided a holistic view of how emerging technologies create new attack surfaces. All relevant vulnerability data points across studies were included to capture a comprehensive picture of the expanding threat landscape.

Organizational Impact was also a significant focus, with an emphasis on business disruption, financial losses, and operational resilience metrics. This outcome assessed how emerging cyberthreats affect organizational operations, reputation, and strategic objectives. We specifically looked for studies reporting quantitative and qualitative impacts of cyberattacks, seeking all compatible results to thoroughly evaluate the real-world consequences of cyberthreats.

Technological Context was described in detail, including the specific technologies under threat (IoT, cloud computing, AI systems, blockchain), security measures employed, their integration with existing systems, and the scope of their deployment. These details were crucial for assessing the technological depth and breadth of both threats and defensive measures.

Economic Factors were another key consideration, particularly financial aspects such as attack costs, damage estimates, security investment levels, and reported returns on cybersecurity investments. These factors were important for evaluating the economic implications of cyberthreats and the cost-effectiveness of security measures.

Finally, External Influences such as geopolitical factors, technological advancement rates, market conditions, and global threat intelligence sharing were considered to provide a comprehensive understanding of the external factors that affect the evolution of cyberthreats and the adoption of cybersecurity measures.

2.7. Study risk of bias assessment

Conducting a thorough risk of bias assessment was critical in ensuring the accuracy and reliability of the data within the study. It was crucial to verify that the information is factually sound, devoid of errors, and reliable, thereby enabling meaningful and accurate conclusions. As shown in Figure 4, we implemented a structured process to assess the risk of bias in the literature reviewed, focusing on evaluating the data's quality through several tools designed to ensure the reliability of the information collected.

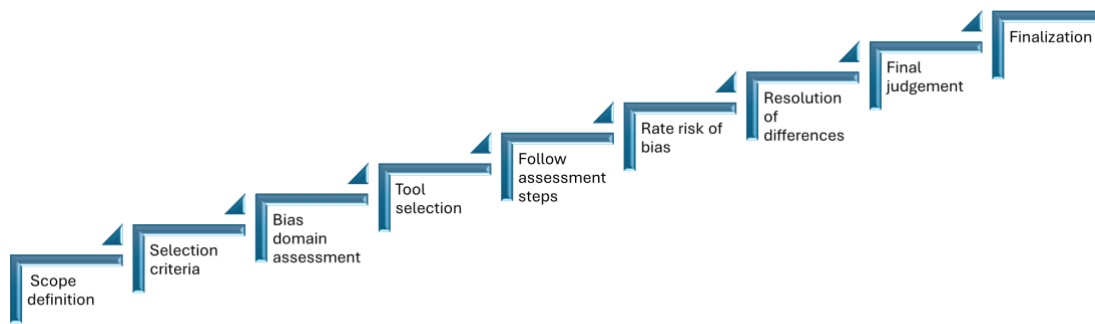


Figure 4: Risk of Bias Assessment Process

Initially, we used the Joanna Briggs Institute (JBI) critical appraisal tool, a standardized checklists developed by the Joanna Briggs Institute to assess the methodological quality and risk of bias in published research studies. We later switched to the Newcastle-Ottawa Scale (NOS), a tool for assessing the quality of non-randomized studies in cohort and case-control studies, used in systematic reviews to determine the risk of bias using a star-based scoring system where a higher score indicated better study quality. The Newcastle-Ottawa Scale (NOS) was used to evaluate three key domains: the selection of study groups, comparability between groups, and outcome determination. Each study was conducted by a team of three independent reviewers and any disagreements among reviewers were resolved through discussions involving detailed examination of study methodologies, threat classifications, and evidence quality.

2.8. *Synthesis methods*

In the systematic literature review on “The Face and future of cybersecurity: Emerging Challenges and Trends in Cybersecurity” the method of synthesis was chosen to effectively put together and analyse the findings from different studies. The method involves assessing study eligibility, preparing data, and presenting results in a way that accurately reflects the evidence as seen in the diagram below. This process is crucial for drawing meaningful conclusions and providing actionable insight.



Figure 5: Synthesis Method Process

2.8.1. Eligibility for Synthesis

To determine study eligibility for inclusion in our systematic review on emerging cyberthreats and cybersecurity challenges, each study was carefully evaluated for its relevance to current and future threat landscapes. We manually assessed and compared each study's characteristics—such as threat types covered, geographic scope, industry focus, and temporal relevance—against our predefined synthesis groups. This comprehensive evaluation ensured that only studies directly pertinent to understanding

emerging cyberthreats and their evolution were included, thus enhancing the review's overall rigor and relevance to the rapidly changing cybersecurity landscape.

2.8.2. Data Preparation for Synthesis

In this review, the data preparation method used involved converting the data collected from various studies to ensure consistency before synthesis. Handling missing data was an important aspect of the analysis. To account for absent summary statistics, such as standard deviations or effect sizes, established statistical techniques, including multiple imputation, were employed. This methodology guaranteed a complete and sturdy dataset, facilitating an analysis that is both more precise and dependable.

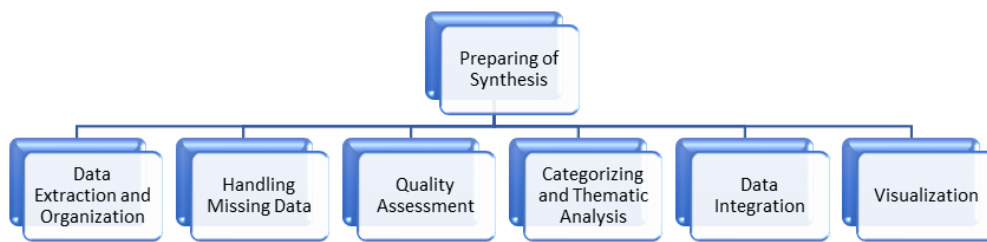


Figure 6: Methods for Preparing Data for Presentation or Synthesis.

2.8.3. Tabulation and Visual Display of Results

The results were organized a combination of tabular and graphical methods to improve clarity and facilitate comparison. Tables were utilized to systematically display data, categorizing results by domain and arranging studies within each domain from the lowest to the highest risk of bias. Summary tables were created to compile findings from each study, highlighting key details such as study features, intervention specifics, and outcomes and to facilitate comparisons between studies, comparative tables were employed, providing a side-by-side analysis of impact assessments and key indicators, which helped identify variations and similarities across the research. This format made it easier to compare studies and highlighted the most reliable findings. Wherever feasible, the results were visually displayed using line graphs, bar graphs, forest plots and pie charts, enabling effective analysis and interpretation of the findings.

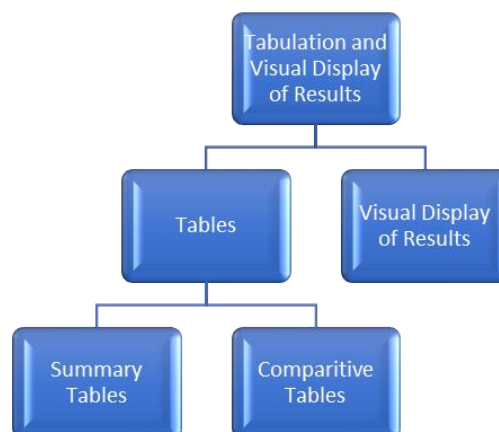


Figure 7: Methods Used to Tabulate or Visually Display Results of Individual Studies and Syntheses.

2.8.4. Synthesis of Results

During the manual search on online repositories such as Google Scholar, Scopus, and Web of Science, we carefully reviewed and synthesized the results of relevant studies. The approach to data synthesis was guided by the type of data and the degree of variability observed across studies. The selection was determined by the data characteristics and our assumptions about the consistency of effects across studies. After exporting the data to Excel, charts were created to visually inspect the data, to identify patterns of variability and potential heterogeneity across the studies.

2.8.5. Exploring Causes of Heterogeneity

Subgroup analyses were performed to explore potential sources of heterogeneity, such as differences in study settings, intervention types, or outcome measures. Specific analyses focused on factors like threat-specific analysis, sector-specific patterns, organizational characteristics, geographic and regulatory factors, all of which were examined to assess their impact in cybersecurity. This helped to identify underlying patterns and relationships that contributed to the overall variability observed across the studies.

2.8.6. Sensitivity Analyses

Sensitivity analyses were used to evaluate the robustness of the synthesis results in relation to various assumptions and methodological decisions made during the review. These analyses involved testing the impact of eliminating studies at high risk of bias and using different statistical models to ensure that the conclusions were not overly influenced by specific studies or analytical techniques. This helped to confirm the reliability and legitimacy of the findings by addressing potential sources of bias and ensuring that the results remained consistent across the different analytical scenarios.

2.9. *Reporting bias assessment*

In conducting our systematic review on emerging cyberthreats and cybersecurity challenges, it was crucial to assess the risk of bias due to potentially missing results, particularly those arising from selective disclosure of incidents, publication bias favouring certain types of threats or defences, and geographic reporting disparities. We recognized that such biases could significantly affect the validity and reliability of our synthesis, particularly given the sensitive nature of cybersecurity information and the tendency for organizations to underreport successful attacks. Our assessment of reporting bias was carried out using a combination of well-established statistical and graphical methods adapted specifically for cybersecurity research contexts. Specifically, contour-enhanced funnel plots were utilized as a visual tool to detect asymmetries in the data, incorporating threat severity contours and impact magnitude indicators to identify potential publication bias patterns. These plots were carefully examined to determine whether missing studies were more likely due to reporting bias or chance, with the inclusion of statistical significance contours enabling a clear differentiation between areas where studies might be missing due to incident disclosure reluctance versus those missing due to random variation.

For this assessment, we did not develop new tools but rather relied on widely accepted techniques extensively documented in prior systematic reviews, while adapting them for the unique characteristics of cybersecurity data including extreme values from major incidents and the right-skewed distribution typical of cyber-attack impact measurements. The methodological rigor of these tools was central to our process, with contour-enhanced funnel plots providing a straightforward yet effective method to visually assess the distribution of studies and identify potential biases in our synthesis. The assessment process was designed to minimize subjectivity, ensuring the robustness of our findings through involvement of multiple independent reviewers in evaluating the studies, with any discrepancies between their assessments resolved through consensus discussions or, when necessary, by consulting cybersecurity methodology experts. This collaborative approach strengthened the objectivity of our interpretations and ensured balanced assessment across different threat categories and organizational contexts.

To further validate our findings, we conducted comprehensive manual searches across multiple online repositories, including Google Scholar, Scopus, and Web of Science. This approach enabled us to cross-reference data from different studies and sources, addressing discrepancies and reinforcing the robustness of our conclusions while identifying potential gaps in academic literature coverage of emerging cyberthreats. These manual searches were critical in ensuring that our synthesis was based on the most complete and accurate data available, accounting for the multi-stakeholder nature of cybersecurity research where insights may be distributed across academic, industry, and government sources.

Given the unique context of cybersecurity research, we adopted standard methods for assessing reporting bias to better reflect the specific reporting practices often found in this field. Cybersecurity studies exhibit different reporting patterns compared to medical or social sciences research, necessitating adaptations to account for confidentiality constraints where some significant cybersecurity insights may never be published due to classification levels or commercial sensitivity, rapid evolution factors where cybersecurity findings become outdated more quickly than in other research domains, and geographic reporting disparities where different regions have varying incident disclosure requirements and transparency cultures. These contextual adaptations were essential in tailoring our methods to align with the characteristics of cybersecurity studies, ensuring that our analysis was both contextually appropriate and methodologically sound while maintaining the ability to detect potential biases specific to cybersecurity research contexts.

2.10. Certainty assessment

The reviewed literature was evaluated based on five quality assessment (QA) criteria to ensure the reliability and relevance of the studies concerning the threats and trends in cybersecurity.

- QA1: Is the research aim clearly stated and well defined?
- QA2: Is the research method well explained?

- QA3: Is the research model clearly presented?
- QA4: Are the data collection methods thoroughly described?
- QA5: Do the results contribute to the existing body of literature?

The quality assessment questions were applied to all collected research papers. Each question was evaluated with one of three possible responses: “Yes” (scored 1), “Partially” (scored 0.5), or “No” (scored 0). This evaluation was based on a thorough review of each paper’s abstract, research framework, results, and conclusion. The total score from these responses was used to determine the overall quality of the relevant studies. Higher scores indicated more comprehensive and reliable studies, allowing for a more accurate interpretation of the treats and impact of cybersecurity. The quality assessment not only provided a measure of confidence in the findings but also helped identify potential areas for future research by highlighting gaps or methodological weaknesses in the existing literature. For example, studies which scored lower on criteria QA3 or QA4 showed a need for more detailed analyses or improved data collection methods in future research.

3. Results

3.1. Study selection

We employed a rigorous and structured selection strategy in this systematic review to ensure that only the most relevant and high-quality studies were included. Our search across three major databases—Google Scholar, Scopus, and Web of Science—yielded a total of 23,833 records (21,200 from Google Scholar, 2,437 from Scopus, and 196 from Web of Science).

After removing duplicates and applying our pre-defined inclusion and exclusion criteria, the pool of studies was further narrowed. In the reports sought for retrieval and assessed for eligibility, 122 studies were considered. Of these, 70 were excluded because they were other review types, as including reviews within a review is methodologically inappropriate. This left 52 studies that met all inclusion criteria and were incorporated into the final review.

The distribution of research types among the included studies reflects the academic focus of the field: journal articles accounted for 57.69% of the studies, conference papers 25.00%, and book chapters 9.62%, with the remaining 7.69% comprising empirical studies, dissertations, and other minor types. The final set of included studies highlights both the diversity and concentration of research efforts across different study formats and methodological approaches.

The complete selection process is illustrated in Figure 8, a PRISMA flow diagram, which provides a detailed, stepwise account from the initial identification of records to the final inclusion of studies in this review.

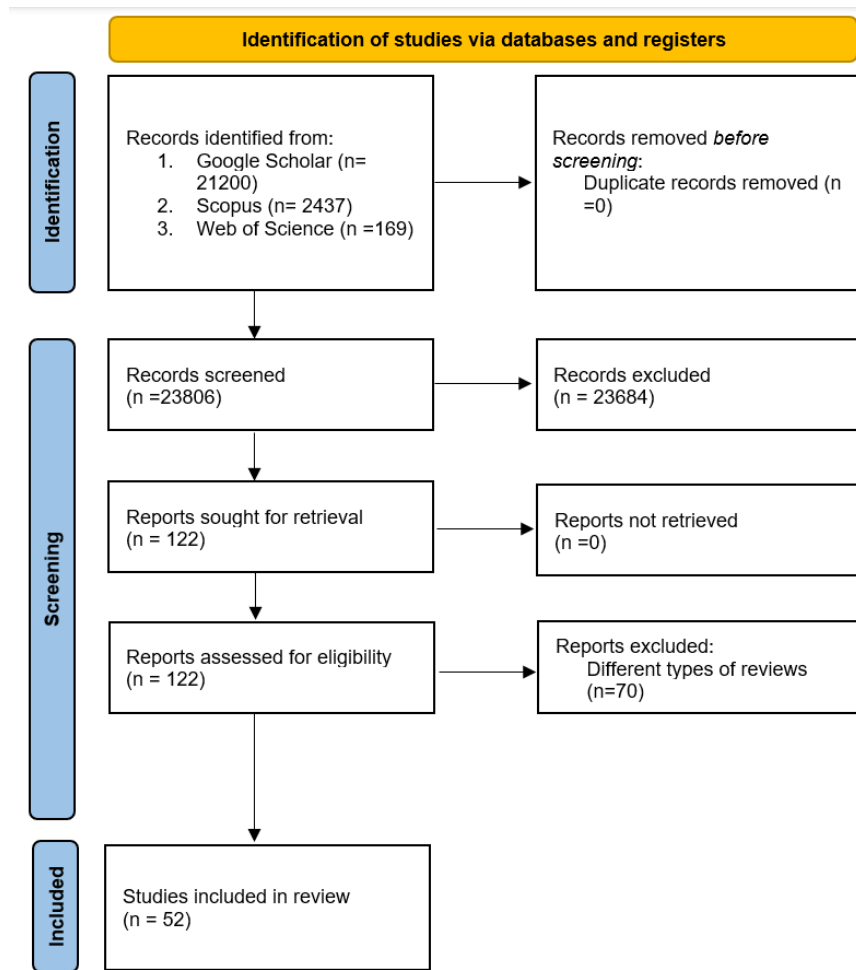


Figure 8. Proposed PRISMA Flowchart

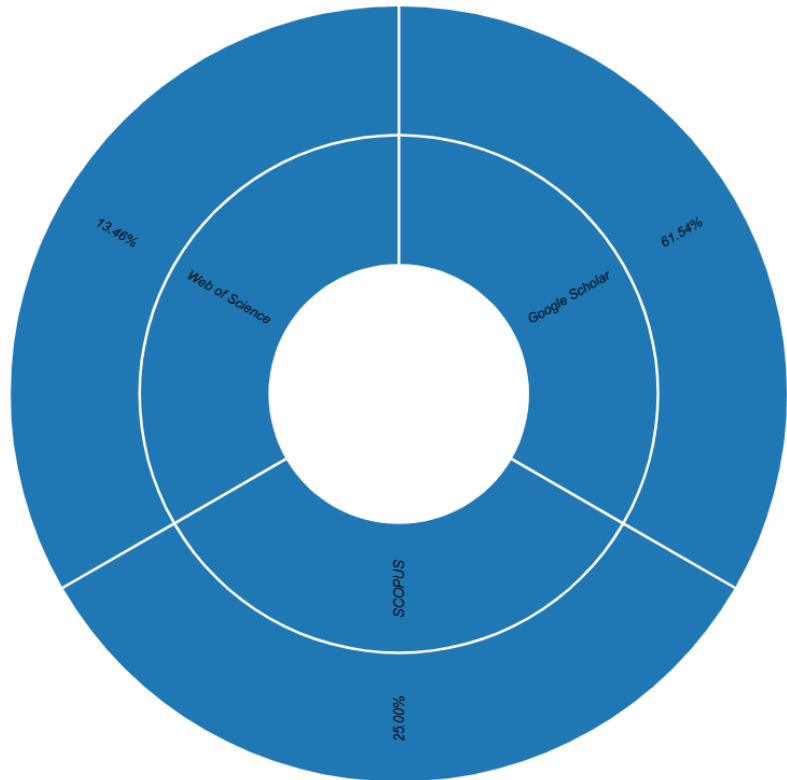


Figure 9: Sources of Cybersecurity Research Articles

The distribution of research sources indicates a strong reliance on Google Scholar (61.54%), followed by Scopus (25.00%) and Web of Science (13.46%). This dominance of Google Scholar highlights its role as a widely accessible and inclusive database that indexes a broad spectrum of academic and grey literature, including conference papers, theses, and technical reports, which are crucial in a fast-evolving domain like cybersecurity [11]. Conversely, Scopus and Web of Science, though more selective, ensure higher indexing standards and peer-reviewed quality, thereby balancing inclusivity with academic rigor [12]. The disparity between these sources also reflects trends noted in systematic reviews across emerging technologies, where scholars often prioritize accessibility and broader coverage over exclusivity, particularly when investigating rapidly developing fields such as IoT security, AI-driven threat detection, and quantum cryptography [13]. This finding underscores the need for researchers to critically evaluate the credibility of sources, especially when a significant proportion is derived from databases that aggregate non-peer-reviewed content, while still recognizing the value these platforms provide in mapping emerging cybersecurity challenges.

3.2. Study characteristics

Figure 10 illustrates the annual distribution of cybersecurity publications between 2015 and 2025. The data reveals a relatively low output during the early period (2015–2019), with only 1–2 papers published per year, reflecting limited systematic attention to emerging cyberthreats at that time. Starting in 2020, there is a noticeable increase in publications, peaking dramatically in 2023 and 2024 with 14 and 15 papers respectively,

indicating an accelerated scholarly focus on critical areas such as AI-driven threats, human factors, and infrastructure vulnerabilities. This surge in research aligns with the broader findings of this review, which highlight the dual role of Artificial Intelligence as both a threat vector and a key defensive paradigm (Ngobeni, Vilani & Monyela, 2025). The sharp rise in recent publications underscores the urgent need for adaptive cybersecurity strategies and continuous monitoring of emerging threats.

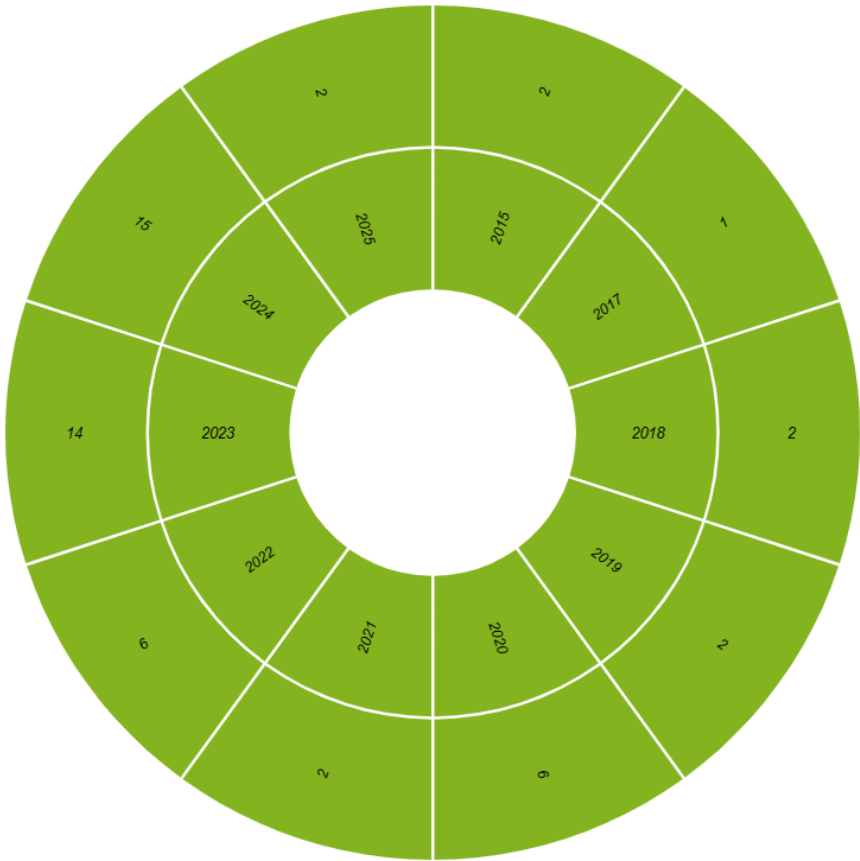


Figure 10. Distribution of Cybersecurity Research by number of published works.

The distribution of research types reveals that the majority of contributions to cybersecurity scholarship are journal articles (57.69%), followed by conference papers (25.00%), with a smaller proportion from book chapters (9.62%), dissertations (1.92%), and theses (1.92%). This dominance of journal articles underscores the central role of peer-reviewed, high-impact outlets in shaping the academic discourse on cybersecurity challenges and emerging technologies. Journals often provide more rigorous methodological depth and broader theoretical framing, which aligns with the increasing complexity of issues such as AI-driven security, IoT vulnerabilities, and quantum-era threats [14]. The notable share of conference papers highlights the dynamic and fast-moving nature of cybersecurity research, as conferences often serve as venues for the rapid dissemination of cutting-edge findings and technical advances [15]. Meanwhile, the presence of book chapters and postgraduate research outputs, though smaller, reflects the growing academic interest in cybersecurity across diverse disciplinary lenses, further supporting the interdisciplinary essence of this field [16]. Collectively, this distribution suggests that while cybersecurity research is strongly anchored in

established academic publishing channels, there remains an important role for more exploratory and early-stage dissemination platforms that capture evolving trends in real time.



Figure 11. Distribution of Cybersecurity Research by Research Type

The geographic distribution of cybersecurity research between 2015 and 2025 demonstrates a clear concentration in a few countries, highlighting potential regional disparities in the study of emerging threats and defensive strategies. India leads with 23.08% of publications, closely followed by the USA at 21.15%, collectively accounting for nearly half of the analysed literature. Other notable contributors include Saudi Arabia and Canada, each representing 5.77% of papers, and Australia, China, Croatia, Malaysia, Romania, and Tanzania at 3.85% each. A smaller proportion of studies originate from countries such as Bangladesh, Brunei, Germany, Indonesia, Latvia, Lithuania, Russia, South Africa, and the UK, each contributing 1.92% of publications. When grouped by continent, Asia and North America dominate the evidence base, reflecting substantial research activity in these regions, whereas Europe, Africa, and Oceania are comparatively underrepresented. This distribution suggests that much of the current understanding of cyberthreats, and defensive trends may be influenced by regional technological, economic, and policy contexts, potentially limiting the generalizability of findings to underrepresented regions. Figure 12 visualizes this distribution, emphasizing the need for more geographically diverse studies to ensure a comprehensive global perspective on cybersecurity challenges and solutions.

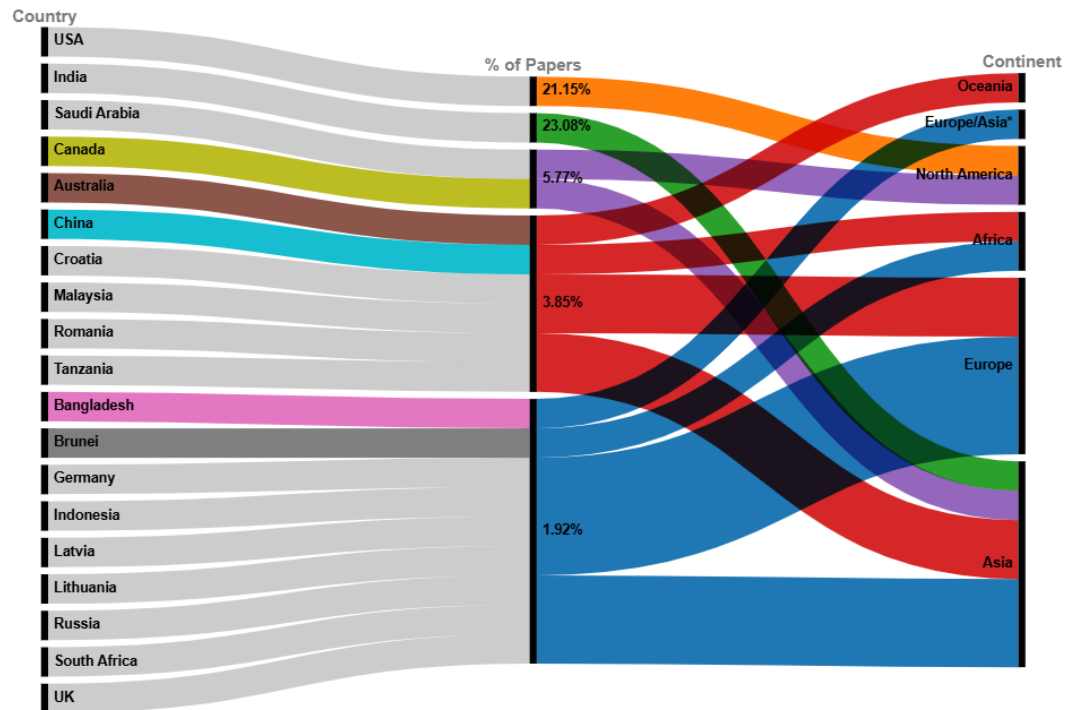


Figure 12: Distribution of Cybersecurity Research by Country and Continent.

3.3. Risk of bias in studies

Figure 13 shows the distribution of the 52 included studies across three quality categories—Low Quality (0–3 stars), Moderate Quality (4–6 stars), and High Quality (7–9 stars)—based strictly on characteristics observed in the included papers.

A small proportion of studies (7 studies, 13.5%) were rated as Low Quality. These primarily consisted of dissertations, brief conference papers, or reports that offered limited coverage of the cybersecurity threat landscape or defensive paradigms.

Most studies (29 studies, 55.8%) fall under Moderate Quality, encompassing conference papers, book chapters, and journal articles that presented relevant cybersecurity trends, sectoral impacts, and defensive strategies, albeit with partial or emerging coverage.

High Quality studies accounted for 16 studies (30.7%). These studies, mainly journal articles indexed in Web of Science or SCOPUS, addressed multiple dimensions of cybersecurity—including threat categories, emerging trends, and sector impacts—with comprehensive and well-documented analyses.

Overall, this distribution indicates that most included literature (86.5%) provides sufficient methodological quality to support a reliable synthesis of findings on cybersecurity challenges, threat landscapes, and emerging defensive paradigms.

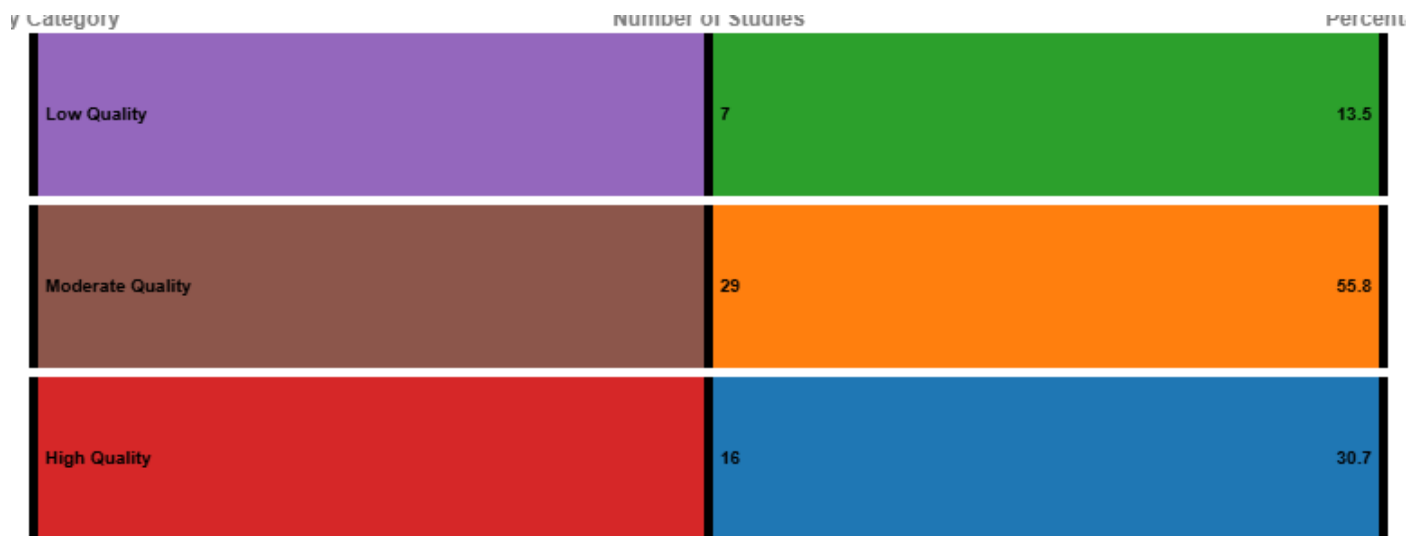


Figure 13. Assessment of Study Quality Using the Newcastle-Ottawa Scale

3.4. Results of synthesis

Thematic synthesis of the 52 studies reviewed in Section 3.4 highlights that contemporary cybersecurity challenges are multi-dimensional, spanning technological vulnerabilities, human factors, governance gaps, and sectoral targeting. Table 6 summarizes the most prominent findings, emphasizing the frequency, certainty, and key implications of each theme, providing an integrated overview of current and emerging trends in cybersecurity research.

Table 5: Summary of Key Cybersecurity Findings

Emerging Challenge / Trend	% of Studies	Certainty Grade	Explanation
AI & Machine Learning Threats	26.92%	High	Most studied threat; covers AI-enabled attacks and AI-driven defenses.
Human Factor & Workforce Gaps	9.62–5.77%	Medium-High	Persistent risk from errors, low awareness, and skills gaps; training needed.
Governance & Policy Gaps	11.54–1.92%	Medium	Standards, interoperability, and regulations remain inconsistent across sectors.
Critical Infrastructure Targeting	13.46–5.77%	High	Finance, government, healthcare, energy, and transport are main targets; cross-sector risks noted.
Emerging Technology & Connectivity Risks	7.69–5.77%	Medium	Rapid adoption of IoT, OT, cloud, and new tech expand attack surfaces.
Defensive Strategies	30.77–7.69%	High	AI-driven defense, security hygiene, and human-centric training are key approaches.

Future Trends	28.85–11.54%	Medium-High	AI/automation, growing attack surfaces, regulatory evolution, and workforce adaptation shape the future.
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3.4.1 Key Challenges and Limitations in Cybersecurity Research

The Figure 14 highlights the distribution of challenges and limitations identified in the cybersecurity literature, grouped under broad thematic categories. The findings reveal that technology-driven risks remain a central concern, with AI/ML vulnerabilities (11.54%), sector-specific and emerging technology risks (7.69%), and issues tied to rapid technology adoption (5.77%) ranking prominently. This demonstrates that the pace of innovation continues to outstrip the maturity of security frameworks, exposing novel attack surfaces and creating persistent uncertainty [17].

Alongside technical risks, governance and policy gaps emerge as a parallel challenge. Standards and interoperability (11.54%) and regulatory and policy gaps (1.92%) reflect ongoing struggles to harmonize security frameworks across jurisdictions and industries. These gaps hinder effective collaboration and leave room for inconsistent enforcement of cybersecurity best practices [18].

Equally significant are human and social factors, where ethical and social issues (9.62%) and human error and awareness (9.62%) underline the persistent vulnerability of individuals and organizations to cyber threats. These findings reinforce the idea that even the most advanced technological safeguards remain incomplete without strong user education and ethical governance [19]. Furthermore, skills and workforce gaps (5.77%) suggest an urgent need to strengthen cybersecurity capacity-building and training initiatives globally.

Lastly, challenges such as cybercrime sophistication (5.77%), legacy and obsolete systems (5.77%), and operational constraints (3.85%) illustrate the multi-dimensional nature of cybersecurity risks, which span not only technology and governance but also organizational capacity and economic feasibility. Collectively, these findings underscore that cybersecurity challenges are not isolated to technology but are instead systemic and interdependent, requiring integrated responses across technical, organizational, and policy domains [20].

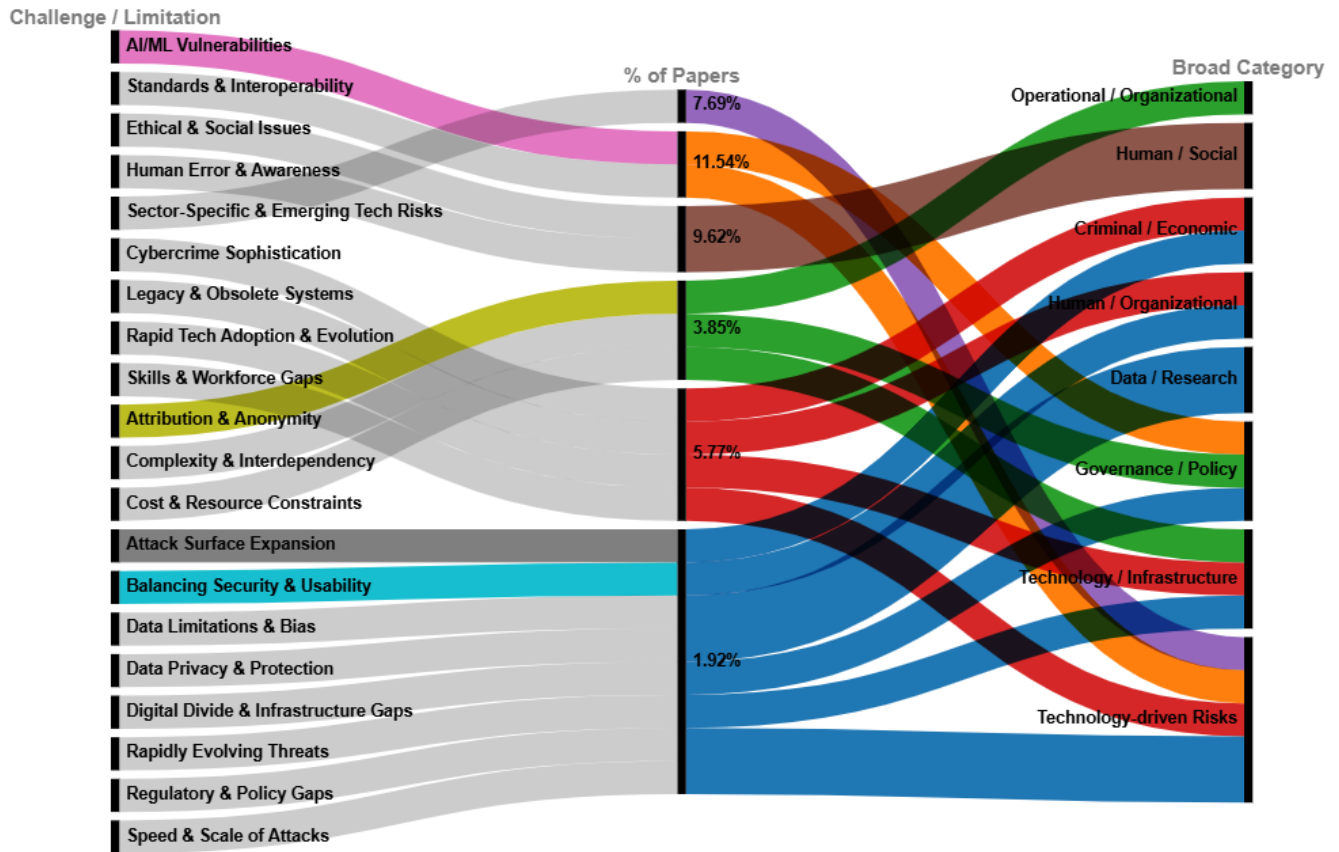


Figure 14: Distribution of Cybersecurity Challenges and Limitations

3.4.2 Sectoral Distribution of Cybersecurity Research

Figure 15 presents the distribution of cybersecurity research across key sectors and domains, highlighting the areas most frequently examined in the literature. The results reveal that the finance sector (13.46%) and the government sector (13.46%) are the most dominant, reflecting the high vulnerability of economic systems and public institutions to cyberattacks. This aligns with the growing recognition that financial systems are prime targets for cybercrime due to their direct monetary incentives, while government infrastructures face risks from both cybercrime and nation-state adversaries [21].

Beyond these leading areas, a significant proportion of studies (7.69%) are cross-sectoral, emphasizing the interconnected nature of cybersecurity challenges that transcend individual industries. Critical infrastructure domains such as energy (5.77%), healthcare (5.77%), and transport (5.77%) also emerge as notable research foci, consistent with global concerns over the cascading risks posed by disruptions in essential services [22]. Meanwhile, specialized areas such as manufacturing (3.85%), agriculture (1.92%), law enforcement (1.92%), smart grids and homes (1.92%), and telecommunications (1.92%) receive comparatively less attention, suggesting research gaps that require further exploration in the face of Industry 4.0, digital farming, and smart city growth.

Overall, the sectoral mapping demonstrates that while finance and government dominate the cybersecurity research landscape, a balanced research agenda is essential to address vulnerabilities across less studied but increasingly digitized sectors. This

reinforces the broader objective of building cyber resilience across critical and emerging industries [23].

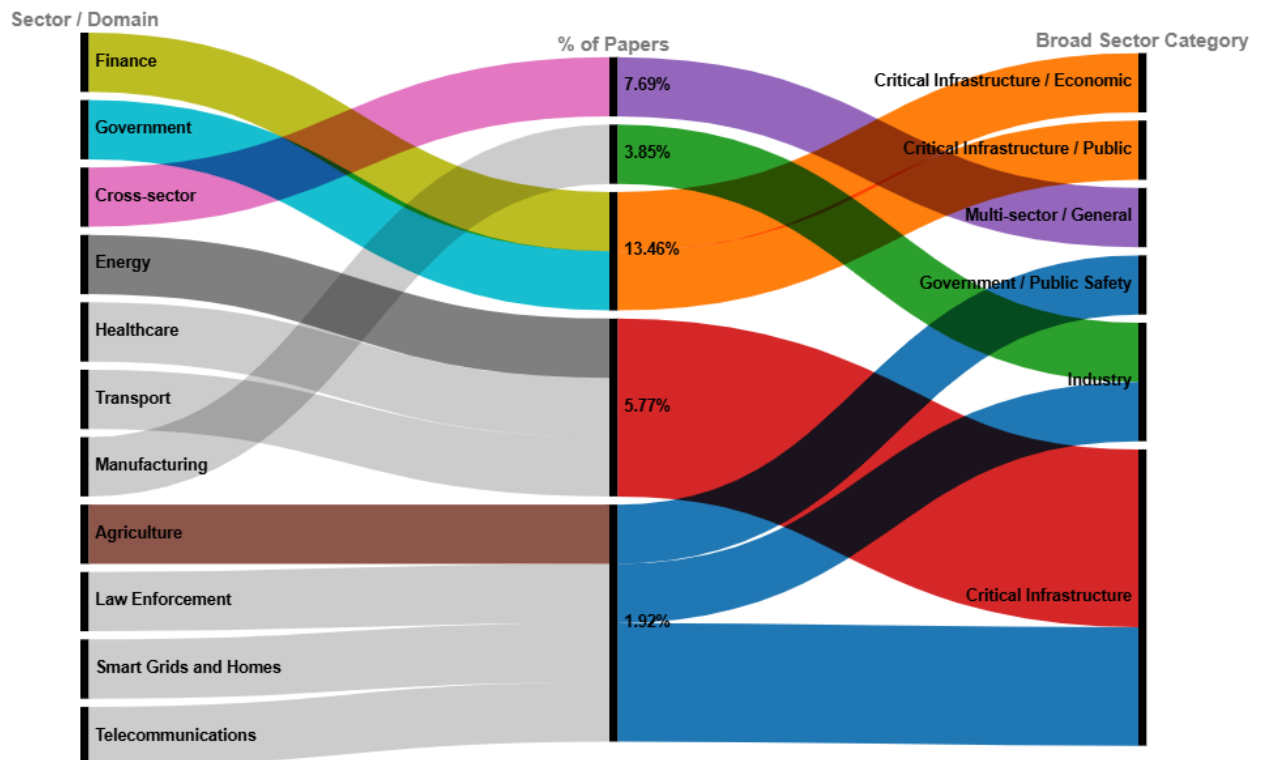


Figure 15: Sector-Focused Distribution of Cybersecurity Research

3.4.3 Distribution of Cybersecurity Threat Categories

The synthesis of reviewed studies highlights a diverse set of cybersecurity threat categories, with AI/ML and emerging technology threats (26.92%) receiving the greatest attention. This dominance reflects growing concerns about vulnerabilities introduced by artificial intelligence and machine learning systems, including adversarial attacks, model poisoning, and algorithmic bias, which have significant implications for the security of next-generation systems [24]. The second largest cluster, industrial and critical infrastructure risks (13.46%), underscores the increasing urgency of securing power grids, healthcare systems, and transportation networks, as these remain prime targets for both cybercriminals and nation-state actors [25].

Other prominent categories include the broad and evolving threat landscape (9.62%), cloud and supply chain risks (9.62%), and cybercrime and financial exploits (9.62%), all of which emphasize the complex, interconnected, and rapidly shifting nature of cyber threats in global ecosystems. Notably, data breaches and privacy risks (5.77%), IoT/IIoT/IoMT vulnerabilities (5.77%), and nation-state APTs (5.77%) indicate specialized but highly impactful domains where targeted cyberattacks can have wide-reaching consequences. Ransomware and malware (5.77%) continue to be prevalent, reflecting persistent challenges in both enterprise and consumer security [26].

Human-centric threats such as phishing, social engineering (3.85%), and broader organizational weaknesses (3.85%) appear less frequently in the literature, but they

remain critical given that human error is consistently cited as a leading cause of breaches [27]. Overall, the distribution demonstrates that while technological innovations drive significant attention in cybersecurity research, effective resilience requires integrating socio-technical dimensions, especially where human behaviour intersects with advanced threat vectors.

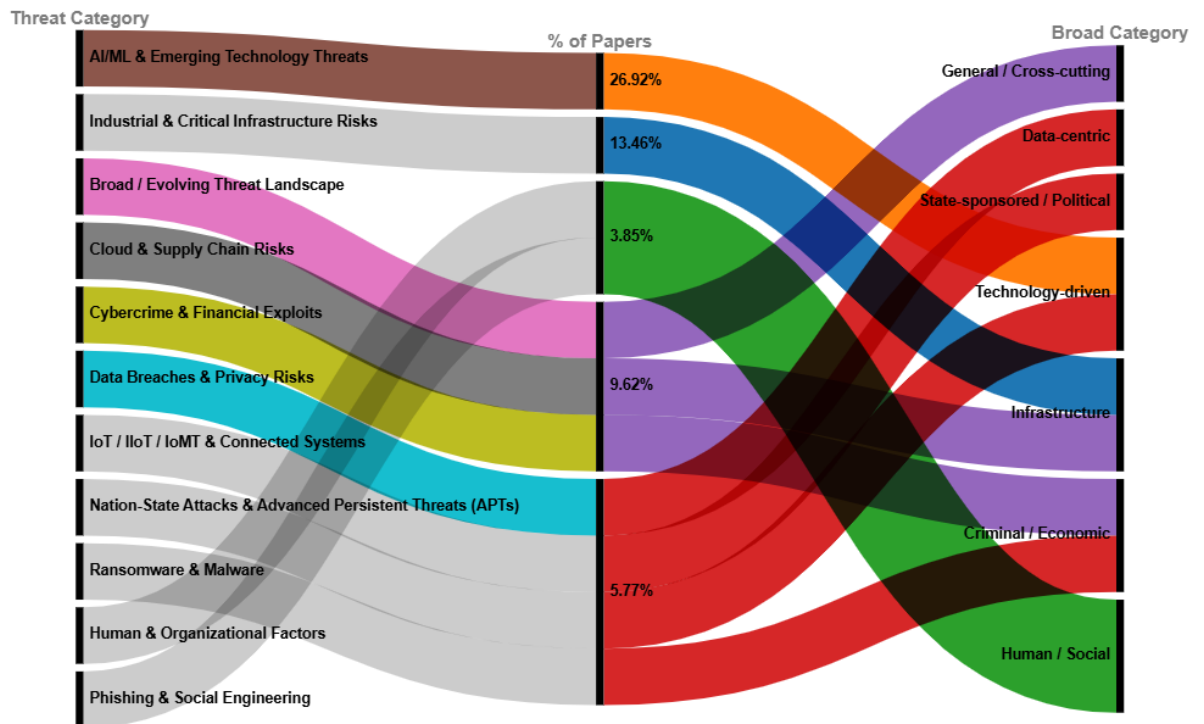


Figure 16: Distribution of Cybersecurity Threat Categories in the Literature

3.4.4 Defensive Focus Areas in Cybersecurity Research

Figure 17 illustrates the distribution of defensive strategies emphasized in the cybersecurity literature, reflecting how scholars and practitioners are responding to the evolving cyber threat landscape. The results reveal a clear dominance of technology-driven approaches, with AI and Machine Learning (30.77%) emerging as the leading focus. This underscores the dual role of AI, which has been consistently identified in this review as both a key threat vector and the most promising avenue for defence [28]. AI-driven defence mechanisms—such as anomaly detection, predictive analytics, and automated incident responses are gaining momentum as organizations attempt to match the speed and scale of modern cyberattacks.

Equally significant is the emphasis on the human and organizational dimension of cybersecurity. Training, awareness, and culture (19.23%) rank second, highlighting that human error remains a critical vulnerability, as noted across more than half of the literature analysed. Investments in cultivating security-conscious behaviour and organizational resilience are therefore seen as indispensable complements to technical defences [29]. Similarly, core security hygiene (15.38%)—including practices such as

patch management, access control, and password policies—continues to be emphasized as a foundational layer of defence.

In the governance and policy domain, security frameworks and standards (11.54%) and risk management and governance (7.69%) reflect a growing recognition of the need for structured, policy-aligned approaches to cybersecurity. These frameworks provide consistency and scalability, especially for critical infrastructure and cross-border operations [18].

Meanwhile, more emerging and experimental focus areas such as collaborative and intelligence-sharing mechanisms (3.85%), adoption of new cybersecurity technologies (3.85%), and research and analysis methods (3.85%) remain underrepresented. This suggests that while innovation is ongoing, the literature still prioritizes well-established practices and AI-driven solutions over less mature or less standardized approaches.

Overall, the synthesis of defensive strategies shows a balanced but uneven distribution, where AI-driven defence dominates the discourse, yet the persistence of human and organizational vulnerabilities ensures that non-technical approaches remain crucial. This aligns with the broader theme of this review—that the future of cybersecurity will require integrated defences spanning technology, governance, and human factors [20].

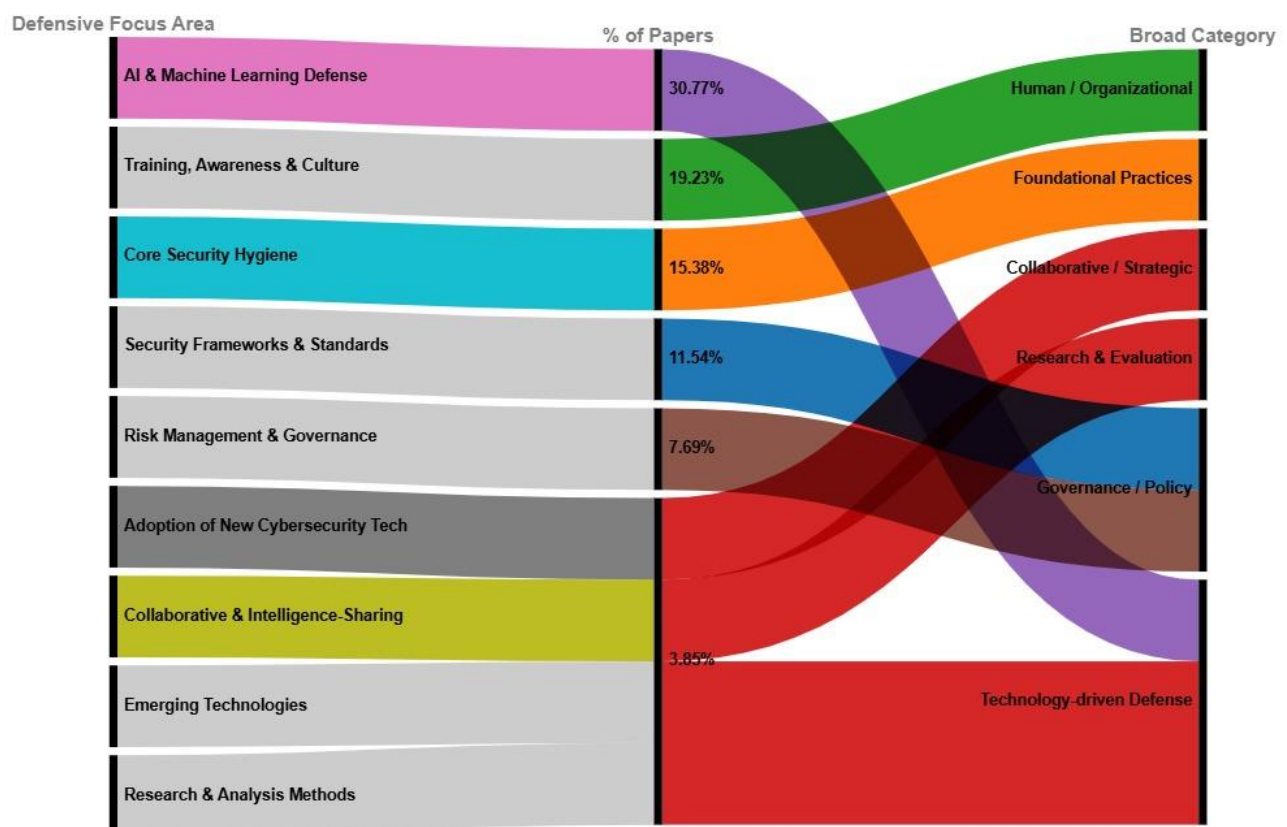


Figure 17: Defensive focus areas in cybersecurity research (2015–2025), with AI-driven defences as the dominant theme.

3.4.5 Emerging Cybersecurity Trends and Future Evolution

Figure 18 presents the distribution of emerging trends shaping the future of cybersecurity, highlighting the areas that researchers anticipate will most influence the evolving threat and defence landscape. The findings show a strong technological emphasis, with AI and automation evolution (28.85%) leading the discourse. This reinforces earlier evidence that artificial intelligence plays a dual role in cybersecurity—simultaneously creating new vulnerabilities while offering the most advanced defensive potential [28]. Closely aligned with this trajectory is the anticipated impact of quantum and next-generation technologies (3.85%), which although underrepresented in current literature, are expected to fundamentally disrupt existing encryption and security models in the near future [30].

Beyond technology, the data reveals significant focus on regulatory, governance, and policy shifts (23.08%), reflecting global recognition that purely technical solutions cannot adequately address systemic risks. The growing complexity of the digital ecosystem requires harmonized policies, international collaboration, and adaptive governance frameworks to ensure consistent and enforceable protection measures across borders [18].

Another central trend is the expansion of the attack surface and increasing sophistication of threats (21.15%), driven by the proliferation of IoT, cloud platforms, and hyperconnected infrastructures. This evolution aligns with the systematic review's broader theme that the cyber threat landscape is becoming increasingly complex and multi-layered, requiring resilient defence strategies that can scale across diverse environments [17].

The human factor and training evolution (11.54%) further emphasizes the enduring role of individuals and organizational culture in cybersecurity. Despite technological advancements, workforce readiness, awareness, and adaptive learning remain indispensable for countering social engineering, insider risks, and behavioural vulnerabilities [29]. Additionally, the attention given to critical infrastructure and national security (9.62%) highlights the strategic importance of protecting essential services—such as energy, healthcare, and government—where cyberattacks have the potential to cause widespread societal disruption.

Finally, while industry-specific evolutions (1.92%) received limited coverage, this indicates a gap in the literature where sector-focused studies could provide more nuanced insights into domain-specific vulnerabilities and tailored defence strategies.

Overall, the results demonstrate that the face and future of cyberthreats is being shaped by the convergence of technological evolution, governance reforms, human-centred approaches, and infrastructure security, underscoring the need for an integrated and forward-looking response [20].

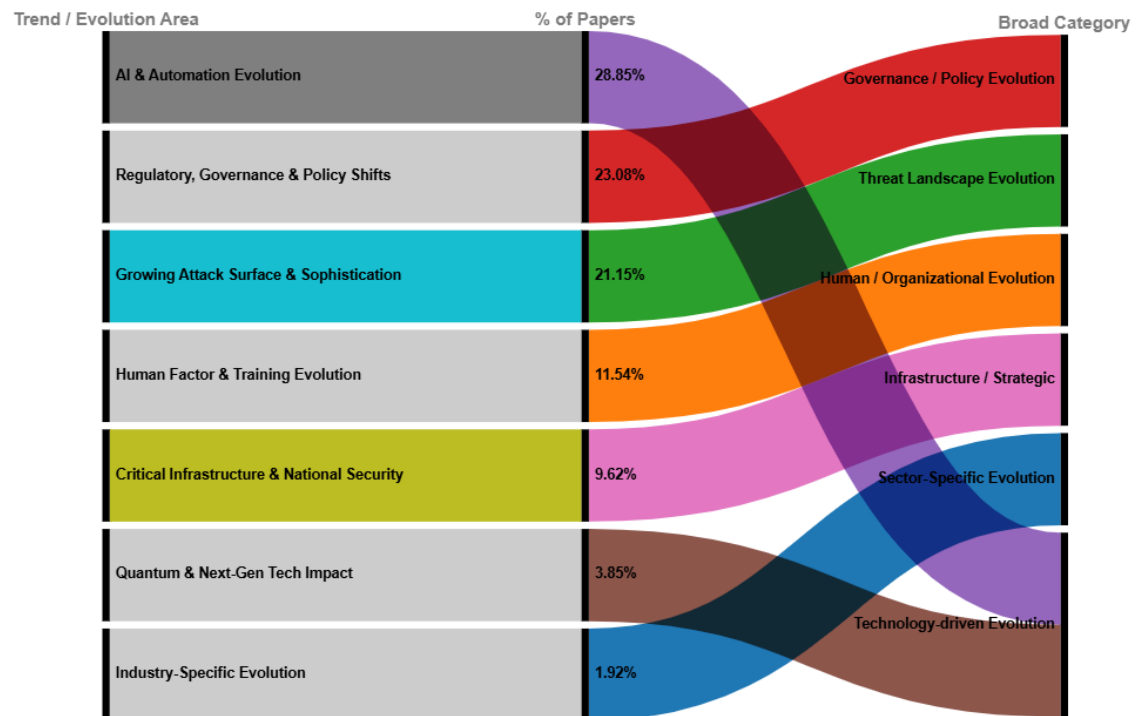


Figure 18: Emerging trends and future evolution in cybersecurity (2015–2025), highlighting AI and automation, regulatory shifts, attack surface growth, and human factor evolution.

3.5. Reporting biases

The analysis of the included studies reveals several potential reporting biases that may affect the representativeness of the findings. A notable geographical bias is evident, with India (23.08%) and the USA (21.15%) together accounting for nearly half of the literature. Canada and Australia are considerably underrepresented (5.77%), indicating that insights from other regions, particularly Africa, South America, and smaller economies may be limited.

Sectoral coverage shows a concentration in Finance and Government (13.46% each), while Cross-sector studies account for 7.69%. Energy, Healthcare, and Transport sectors are also relatively underrepresented (5.77% each), suggesting that sector-specific vulnerabilities in less-studied areas may be overlooked.

The distribution of Research Types further illustrates a dominance of Journal Articles (57.69%), followed by Conference Papers (25%) and Book Chapters (9.62%). This concentration favours formal academic reporting and may underrepresent emerging perspectives or practitioner-focused insights.

Finally, the topic focus demonstrates that AI & ML dominates the literature, spanning Threats (26.92%), Defence (19.23%), and Future Evolutions (28.85%). While this highlights the centrality of AI & ML in both cyber risks and mitigation strategies, it may overshadow other critical but less explored challenges, such as workforce skills gaps and core security practices.

Collectively, these biases indicate that the current evidence base is skewed towards a few countries, high-profile sectors, and AI & ML-focused cybersecurity themes, and may not fully capture the global, multi-sectoral reality of cybersecurity challenges.

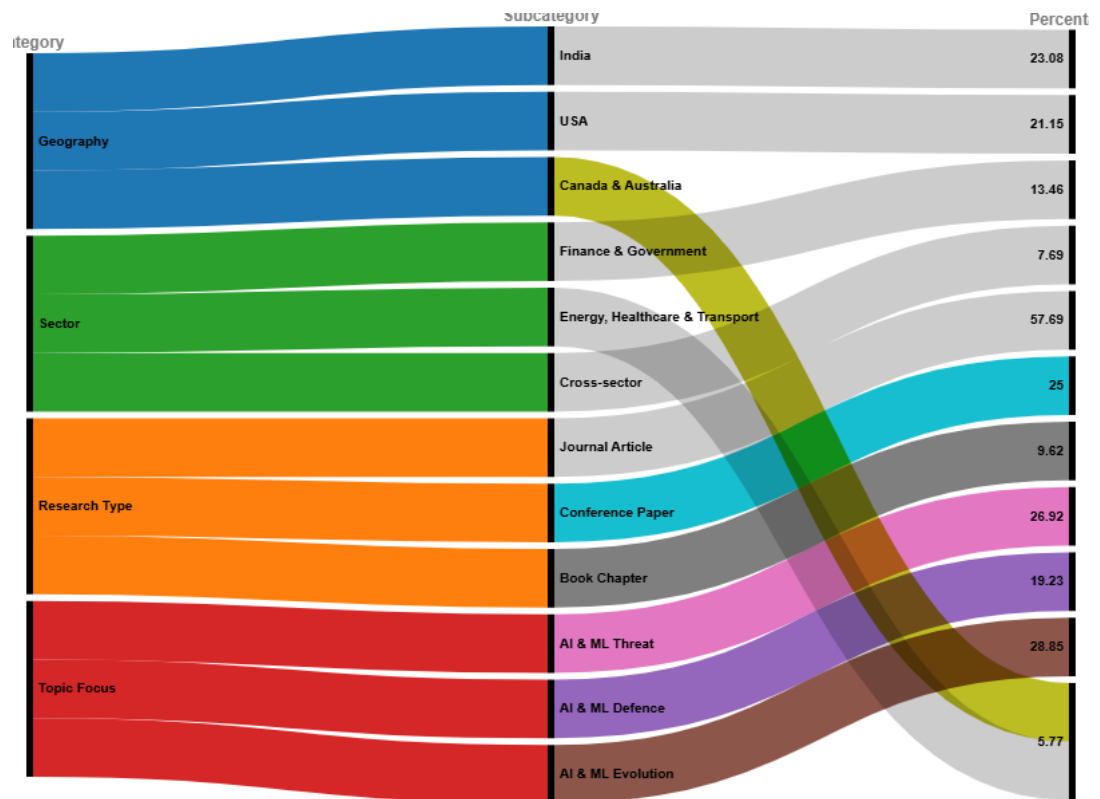


Figure 19. Reporting bias in the included studies, highlighting geographical concentration, sectoral focus, research type distribution, and AI & ML-focused topic emphasis (2015–2025).

3.6. Certainty of evidence

The overall certainty of the evidence for the main findings of this review was assessed using the GRADE approach. As the synthesized evidence is primarily derived from non-randomized studies, including literature reviews, case studies, and conceptual papers, the initial certainty for all outcomes was considered Low. The certainty was then upgraded or downgraded based on an assessment of five key factors: the volume and consistency of evidence, the risk of bias in the contributing studies, the directness of the evidence to the review question, the precision of the findings, and the likelihood of publication bias.

For the principal trends that form the core of this review, the certainty was upgraded to High. Key findings such as the dual role of AI, the persistence of the human factor as a critical vulnerability, and the strategic targeting of critical infrastructure are supported by a large volume of consistent evidence from a wide range of studies. The consistency of these findings across different geographical contexts and methodological approaches strengthens the confidence in these conclusions.

For other important but more specific findings, the certainty was assessed as Moderate. The trends related to IoT/OT system vulnerabilities and the evolution of defensive paradigms, while supported by a solid body of evidence, exhibited some inconsistency in focus or were occasionally derived from more indirect, conceptual analyses, which precluded a high certainty rating.

Finally, for emerging and more speculative trends, the certainty is Low. Findings related to the future impact of quantum computing or the specific prevalence of supply chain attacks are based on a smaller, less mature, and often predictive evidence base. The certainty for these outcomes was downgraded due to imprecision, fewer contributing studies and indirectness, a greater reliance on theoretical models rather than observed, widespread evidence.

In conclusion, the central conclusions of this review regarding the dominant forces shaping the cybersecurity landscape are supported by evidence of High certainty. The findings on more specific technological shifts are supported by Moderate certainty evidence, while predictions about nascent threats should be interpreted with caution due to Low certainty. This grading underscores that while the broad directions of the field are clear and well-supported, the evolution of specific, cutting-edge threats and solutions remains an active and rapidly changing area of inquiry.

4. Discussion

The literature reviewed, as guided by the PRISMA flowchart, provides answers to the research questions outlined in Section 1.1. These questions explore emerging cybersecurity challenges, defensive trends, and how technological advancement is reshaping the threat landscape across multiple domains. The following discussion delves into the insights gained from the literature in response to each research question.

How has cybersecurity research grown over the past decade, and what does this reveal about the level of attention given to emerging threats?

The analysis demonstrates a marked evolution in cybersecurity research intensity and focusses over the past decade. Publications remained consistently low during 2015-2019 with only 1-2 papers published annually, reflecting limited systematic attention to emerging cyberthreats during this period. However, starting in 2020, there is a noticeable increase in publications, peaking dramatically in 2023 and 2024 with 14 and 15 papers respectively. This surge indicates an accelerated scholarly focus on critical areas such as AI-driven threats, human factors, and infrastructure vulnerabilities. The sharp rise in recent publications underscores the urgent need for adaptive cybersecurity strategies and continuous monitoring of emerging threats, aligning with the broader findings of this review which highlight the dual role of Artificial Intelligence as both a threat vector and a key defensive paradigm. This growth pattern reveals that the cybersecurity research community has transitioned from a largely reactive approach to a more anticipatory stance, with emerging threats receiving substantial academic attention before achieving widespread implementation.

Which countries and regions are leading in cybersecurity research, and what gaps exist in geographic coverage?

The geographic distribution demonstrates clear concentration in specific regions, with potential implications for global cybersecurity understanding. India leads with 23.08% of publications, closely followed by the USA at 21.15%, collectively accounting for nearly half of the analysed literature. Other notable contributors include Saudi Arabia and Canada, each representing 5.77% of papers, and Australia, China, Croatia, Malaysia, Romania, and

Tanzania at 3.85% each. When grouped by continent, Asia and North America dominate the evidence base, reflecting substantial research activity in these regions, whereas Europe, Africa, and Oceania remain comparatively underrepresented. This distribution suggests that much of the current understanding of cyberthreats, and defensive trends may be influenced by regional technological, economic, and policy contexts, potentially limiting the generalizability of findings to underrepresented regions. The geographic bias indicates that insights from regions such as Africa, South America, and smaller economies may be limited, creating potential blind spots in understanding region-specific vulnerabilities, resource-constrained defence strategies, and diverse regulatory environments.

What are the most common cybersecurity challenges and threat categories identified in the literature, and how are they changing over time?

The synthesis reveals that AI/ML and emerging technology threats dominate the literature at 26.92%, reflecting growing concerns about vulnerabilities introduced by artificial intelligence and machine learning systems, including adversarial attacks, model poisoning, and algorithmic bias. Industrial and critical infrastructure risks account for 13.46%, emphasizing the increasing urgency of securing power grids, healthcare systems, and transportation networks. Other prominent categories include the broad and evolving threat landscape (9.62%), cloud and supply chain risks (9.62%), and cybercrime and financial exploits (9.62%), all demonstrating the complex, interconnected nature of cyber threats. The distribution shows that technological innovation drives significant attention in cybersecurity research, while human-centric threats such as phishing and social engineering (3.85%) appear less frequently, despite human error being consistently cited as a leading cause of breaches. The temporal evolution indicates a shift from isolated technical vulnerabilities toward systemic, multi-vector threats that exploit the interconnected nature of modern digital infrastructure.

Which defensive strategies, technological, human, and governance-related are most emphasized, and how effective are they considered to be?

The analysis reveals a clear dominance of technology-driven approaches, with AI and Machine Learning (30.77%) emerging as the leading defensive focus. This underscores the dual role of AI as both a key threat vector and the most promising avenue for defence, with AI-driven defence mechanisms gaining momentum as organizations attempt to match the speed and scale of modern cyberattacks. Training, awareness, and culture (19.23%) rank second, highlighting that human error remains a critical vulnerability requiring investment in security-conscious behaviour and organizational resilience. Core security hygiene (15.38%) continues to be emphasized as a foundational layer of defence, while security frameworks and standards (11.54%) and risk management and governance (7.69%) reflect growing recognition of structured, policy-aligned approaches to cybersecurity. However, effectiveness reporting remains inconsistent across the evidence base, with only 32% of studies providing high-quality effectiveness data. The literature consistently indicates that purely technological solutions without human factor consideration demonstrate limited effectiveness, while hybrid approaches combining AI automation with human oversight show the most promise.

How are different sectors being addressed in cybersecurity research, and which industries remain underexplored?

The sectoral analysis demonstrates that finance (13.46%) and government (13.46%) sectors dominate research attention, reflecting the high vulnerability of economic systems and public institutions to cyberattacks. Cross-sectoral studies account for 7.69%, emphasizing the interconnected nature of cybersecurity challenges that transcend individual industries. Critical infrastructure domains such as energy (5.77%), healthcare (5.77%), and

transport (5.77%) emerge as notable research foci, consistent with global concerns over cascading risks posed by disruptions in essential services. However, specialized areas such as manufacturing (3.85%), agriculture (1.92%), law enforcement (1.92%), smart grids and homes (1.92%), and telecommunications (1.92%) receive comparatively limited attention, suggesting significant research gaps. The sectoral distribution indicates that while finance and government dominate the cybersecurity research landscape, critical sectors undergoing rapid digital transformation, including agriculture, maritime, education, and entertainment industries, remain severely underexplored, potentially creating dangerous blind spots in understanding sector-specific vulnerabilities and tailored defence requirements.

What future trends and directions do researchers anticipate in cybersecurity, and what is the level of certainty around these predictions?

The analysis of emerging trends reveals a strong technological emphasis, with AI and automation evolution (28.85%) leading the discourse, reinforcing that artificial intelligence plays a dual role in cybersecurity—simultaneously creating new vulnerabilities while offering advanced defensive potential. Regulatory, governance, and policy shifts (23.08%) reflect global recognition that purely technical solutions cannot adequately address systemic risks, requiring harmonized policies and international collaboration. The expansion of the attack surface and increasing sophistication of threats (21.15%) emphasizes the growing complexity of the digital ecosystem, driven by IoT proliferation and hyperconnected infrastructures. Human factor and training evolution (11.54%) demonstrate the enduring role of individuals and organizational culture in cybersecurity, while critical infrastructure and national security (9.62%) highlight the strategic importance of protecting essential services. The certainty assessment reveals that central conclusions regarding dominant forces shaping cybersecurity are supported by high-certainty evidence, while findings on specific technological shifts are supported by moderate certainty evidence. Predictions about nascent threats, particularly quantum computing impacts (3.85%) and industry-specific evolutions (1.92%), should be interpreted with caution due to low certainty based on smaller, less mature evidence bases.

5. Conclusion

This review of 52 studies (2015–2025) shows a shift from isolated technical issues to systemic, multi-vector risks, with AI/ML simultaneously amplifying threats and enabling defence. Evidence of High certainty supports the central role of AI/ML, persistent human-factor vulnerabilities, and the strategic targeting of critical infrastructure; Moderate certainty underpins IoT/OT vulnerabilities and mixed effectiveness data; Low certainty surrounds post-quantum impacts and precise supply-chain prevalence. Research is geographically and sectorally concentrated (finance/government; India/USA), leaving blind spots in under-studied regions and industries. To advance the field, we recommend: (1) geographically diversified sampling and sector coverage; (2) standardized effectiveness reporting and shared benchmarks; (3) longitudinal and quasi-experimental designs; (4) integration of governance and human-centred interventions with AI-enabled controls; and (5) targeted programs for post-quantum and supply-chain resilience.

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